Lekcja 34- Terragrunt - Skalowalne i suche (DRY) IaC

1.Stwarzamy strukturę infrastructure-live-staging

```
C:.
└──terragrunt-aws-lab
    ├──infrastructure-live
    │   ├──dev
    │   │   ├──app
    │   │   └──vpc
    │   └──staging
    │       ├──app
    │       └──vpc
    └──modules
```

2.Struktura multi-region

```
terragrunt-aws-lab
  infrastructure-live
    dev
      app
      vpc
      region.hcl
    staging
      app
      vpc
      region.hcl
  modules
```

3.Terragrunt-root

```
region.hcl ...\staging        terragrunt.hcl
terragrunt-aws-lab > infrastructure-live > terragrunt.hcl
1   locals {
2     region_vars = read_terragrunt_con
3     aws_region  = local.region_vars.l
4   }
5
6   generate "provider" {
7     path = "provider.tf"
8     if_exists = "overwrite_terragrunt"
9     contents = <<EOF
10  provider "aws" {
11    region = "${local.aws_region}"
12  }
13  EOF
14  }
```

4.Struktura projektu

```
C:.
+---infrastructure-live
|   |   terragrunt.hcl
|   |
|   +---dev
|   |   |   region.hcl
|   |   |
|   |   +---app
|   |   |       terragrunt.hcl
|   |   |
|   |   \---vpc
|   |           terragrunt.hcl
|   |
|   \---staging
|       |   region.hcl
|       |
|       +---app
|       |       terragrunt.hcl
|       |
|       \---vpc
|               terragrunt.hcl
|
\---modules
    +---app
    |       main.tf
    |
    \---vpc
            main.tf
```

5.Terraform wynik1

```
PS C:\Users\user\Desktop\zad1\lesson34\terragrunt-aws-lab\infrastructure-live\dev> terragrunt run --all plan
01:34:44.616 WARN   Using `terragrunt.hcl` as the root of Terragrunt configurations is an anti-pattern, and no longer recommende
d. In a future version of Terragrunt, this will result in an error. You are advised to use a differently named file like `root.h
cl` instead. For more information, see https://docs.terragrunt.com/migrate/migrating-from-root-terragrunt-hcl
01:34:44.698 INFO   - Unit vpc
- Unit app

01:34:52.898 STDOUT [vpc] terraform: Terraform used the selected providers to generate the following execution
01:34:52.898 STDOUT [vpc] terraform: plan. Resource actions are indicated with the following symbols:
01:34:52.899 STDOUT [vpc] terraform:   + create
01:34:52.899 STDOUT [vpc] terraform: Terraform will perform the following actions:
01:34:52.900 STDOUT [vpc] terraform:   # aws_subnet.public will be created
01:34:52.900 STDOUT [vpc] terraform:   + resource "aws_subnet" "public" {
01:34:52.900 STDOUT [vpc] terraform:       + arn                                            = (known after apply)
01:34:52.900 STDOUT [vpc] terraform:       + assign_ipv6_address_on_creation                = false
01:34:52.900 STDOUT [vpc] terraform:       + availability_zone                              = (known after apply)
01:34:52.900 STDOUT [vpc] terraform:       + availability_zone_id                           = (known after apply)
01:34:52.901 STDOUT [vpc] terraform:       + cidr_block                                     = "10.0.1.0/24"
01:34:52.901 STDOUT [vpc] terraform:       + enable_dns64                                   = false
01:34:52.901 STDOUT [vpc] terraform:       + enable_resource_name_dns_a_record_on_launch    = false
01:34:52.901 STDOUT [vpc] terraform:       + enable_resource_name_dns_aaaa_record_on_launch = false
01:34:52.901 STDOUT [vpc] terraform:       + id                                             = (known after apply)
01:34:52.901 STDOUT [vpc] terraform:       + ipv6_cidr_block                                = (known after apply)
01:34:52.901 STDOUT [vpc] terraform:       + ipv6_cidr_block_association_id                 = (known after apply)
01:34:52.901 STDOUT [vpc] terraform:       + ipv6_native                                    = false
01:34:52.901 STDOUT [vpc] terraform:       + map_public_ip_on_launch                        = false
01:34:52.902 STDOUT [vpc] terraform:       + owner_id                                       = (known after apply)
01:34:52.902 STDOUT [vpc] terraform:       + private_dns_hostname_type_on_launch            = (known after apply)
01:34:52.902 STDOUT [vpc] terraform:       + region                                         = "eu-central-1"
01:34:52.902 STDOUT [vpc] terraform:       + tags                                           = {
01:34:52.902 STDOUT [vpc] terraform:           + "Name" = "dev-vpc-public"
01:34:52.903 STDOUT [vpc] terraform:         }
01:34:52.903 STDOUT [vpc] terraform:       + tags_all                                       = {
01:34:52.903 STDOUT [vpc] terraform:           + "Name" = "dev-vpc-public"
01:34:52.903 STDOUT [vpc] terraform:         }
01:34:52.903 STDOUT [vpc] terraform:       + vpc_id                                         = (known after apply)
01:34:52.903 STDOUT [vpc] terraform:     }
01:34:52.903 STDOUT [vpc] terraform:   # aws_vpc.main will be created
01:34:52.903 STDOUT [vpc] terraform:   + resource "aws_vpc" "main" {
01:34:52.903 STDOUT [vpc] terraform:       + arn                                  = (known after apply)
01:34:52.903 STDOUT [vpc] terraform:       + cidr_block                           = "10.0.0.0/16"
01:34:52.903 STDOUT [vpc] terraform:       + default_network_acl_id               = (known after apply)
01:34:52.904 STDOUT [vpc] terraform:       + default_route_table_id               = (known after apply)
01:34:52.904 STDOUT [vpc] terraform:       + default_security_group_id            = (known after apply)
01:34:52.905 STDOUT [vpc] terraform:       + dhcp_options_id                      = (known after apply)
01:34:52.905 STDOUT [vpc] terraform:       + enable_dns_hostnames                 = (known after apply)
01:34:52.907 STDOUT [vpc] terraform:       + enable_dns_support                   = true
01:34:52.909 STDOUT [vpc] terraform:       + enable_network_address_usage_metrics = (known after apply)
```

6.Terraform wynik2

```
01:34:52.910 STDOUT [vpc] terraform:       + id                                   = (known after apply)
01:34:52.911 STDOUT [vpc] terraform:       + instance_tenancy                     = "default"
01:34:52.921 STDOUT [vpc] terraform:       + ipv6_association_id                  = (known after apply)
01:34:52.921 STDOUT [vpc] terraform:       + ipv6_cidr_block                      = (known after apply)
01:34:52.923 STDOUT [vpc] terraform:       + ipv6_cidr_block_network_border_group = (known after apply)
01:34:52.924 STDOUT [vpc] terraform:       + main_route_table_id                  = (known after apply)
01:34:52.924 STDOUT [vpc] terraform:       + owner_id                             = (known after apply)
01:34:52.931 STDOUT [vpc] terraform:       + region                               = "eu-central-1"
01:34:52.935 STDOUT [vpc] terraform:       + tags                                 = {
01:34:52.936 STDOUT [vpc] terraform:           + "Name" = "dev-vpc"
01:34:52.937 STDOUT [vpc] terraform:         }
01:34:52.938 STDOUT [vpc] terraform:       + tags_all                             = {
01:34:52.939 STDOUT [vpc] terraform:           + "Name" = "dev-vpc"
01:34:52.940 STDOUT [vpc] terraform:         }
01:34:52.940 STDOUT [vpc] terraform:     }
01:34:52.942 STDOUT [vpc] terraform: Plan: 2 to add, 0 to change, 0 to destroy.
01:34:52.943 STDOUT [vpc] terraform:
01:34:52.947 STDOUT [vpc] terraform: Changes to Outputs:
01:34:52.948 STDOUT [vpc] terraform:   + subnet_id = (known after apply)
01:34:52.949 STDOUT [vpc] terraform:
01:34:52.950 STDOUT [vpc] terraform: ────────────────────────────────────────────────────────────────────────────────
01:34:52.951 STDOUT [vpc] terraform: Note: You didn't use the -out option to save this plan, so Terraform can't
01:34:52.952 STDOUT [vpc] terraform: guarantee to take exactly these actions if you run "terraform apply" now.
01:34:54.694 WARN   [app] Config C:\Users\user\Desktop\zad1\lesson34\terragrunt-aws-lab\infrastructure-live\dev\vpc\terragrunt.h
cl is a dependency of C:\Users\user\Desktop\zad1\lesson34\terragrunt-aws-lab\infrastructure-live\dev\app\terragrunt.hcl that has
 no outputs, but mock outputs provided and returning those in dependency output.
01:34:54.702 INFO   [app] Downloading Terraform configurations from ..\..\modules\app into .\app\.terragrunt-cache\L0_XAIrspWRGr
4pdaURKYtO3X-Y\oPcMacoiQTbBcQruEA5c4BO_G-g
01:34:54.899 INFO   [app] terraform: Initializing the backend...
01:34:56.478 INFO   [app] terraform: Initializing provider plugins...
01:34:56.478 INFO   [app] terraform: - Reusing previous version of hashicorp/aws from the dependency lock file
01:34:58.850 INFO   [app] terraform: - Using previously-installed hashicorp/aws v6.42.0
01:34:58.851 INFO   [app] terraform: Terraform has been successfully initialized!
01:34:58.853 INFO   [app] terraform:
01:34:58.854 INFO   [app] terraform: You may now begin working with Terraform. Try running "terraform plan" to see
01:34:58.857 INFO   [app] terraform: any changes that are required for your infrastructure. All Terraform commands
01:34:58.858 INFO   [app] terraform: should now work.
01:34:58.859 INFO   [app] terraform: If you ever set or change modules or backend configuration for Terraform,
01:34:58.860 INFO   [app] terraform: rerun this command to reinitialize your working directory. If you forget, other
01:34:58.862 INFO   [app] terraform: commands will detect it and remind you to do so if necessary.
01:35:07.313 STDOUT [app] terraform: Terraform used the selected providers to generate the following execution
01:35:07.313 STDOUT [app] terraform: plan. Resource actions are indicated with the following symbols:
01:35:07.314 STDOUT [app] terraform:   + create
01:35:07.316 STDOUT [app] terraform: Terraform will perform the following actions:
01:35:07.318 STDOUT [app] terraform:   # aws_instance.server will be created
01:35:07.320 STDOUT [app] terraform:   + resource "aws_instance" "server" {
01:35:07.323 STDOUT [app] terraform:       + ami                                  = "ami-12345678"
01:35:07.324 STDOUT [app] terraform:       + arn                                  = (known after apply)
01:35:07.326 STDOUT [app] terraform:       + associate_public_ip_address          = (known after apply)
```

7.Terraform wynik3

```
01:35:07.327 STDOUT [app] terraform:       + availability_zone                    = (known after apply)
01:35:07.327 STDOUT [app] terraform:       + disable_api_stop                     = (known after apply)
01:35:07.328 STDOUT [app] terraform:       + disable_api_termination              = (known after apply)
01:35:07.329 STDOUT [app] terraform:       + ebs_optimized                        = (known after apply)
01:35:07.331 STDOUT [app] terraform:       + enable_primary_ipv6                  = (known after apply)
01:35:07.332 STDOUT [app] terraform:       + force_destroy                        = false
01:35:07.333 STDOUT [app] terraform:       + get_password_data                    = false
01:35:07.335 STDOUT [app] terraform:       + host_id                              = (known after apply)
01:35:07.340 STDOUT [app] terraform:       + host_resource_group_arn              = (known after apply)
01:35:07.341 STDOUT [app] terraform:       + iam_instance_profile                 = (known after apply)
01:35:07.343 STDOUT [app] terraform:       + id                                   = (known after apply)
01:35:07.344 STDOUT [app] terraform:       + instance_initiated_shutdown_behavior = (known after apply)
01:35:07.345 STDOUT [app] terraform:       + instance_lifecycle                   = (known after apply)
01:35:07.346 STDOUT [app] terraform:       + instance_state                       = (known after apply)
01:35:07.346 STDOUT [app] terraform:       + instance_type                        = "t3.micro"
01:35:07.347 STDOUT [app] terraform:       + ipv6_address_count                   = (known after apply)
01:35:07.348 STDOUT [app] terraform:       + ipv6_addresses                       = (known after apply)
01:35:07.349 STDOUT [app] terraform:       + key_name                             = (known after apply)
01:35:07.350 STDOUT [app] terraform:       + monitoring                           = (known after apply)
01:35:07.355 STDOUT [app] terraform:       + outpost_arn                          = (known after apply)
01:35:07.357 STDOUT [app] terraform:       + password_data                        = (known after apply)
01:35:07.358 STDOUT [app] terraform:       + placement_group                      = (known after apply)
01:35:07.359 STDOUT [app] terraform:       + placement_group_id                   = (known after apply)
01:35:07.360 STDOUT [app] terraform:       + placement_partition_number           = (known after apply)
01:35:07.361 STDOUT [app] terraform:       + primary_network_interface_id         = (known after apply)
01:35:07.362 STDOUT [app] terraform:       + private_dns                          = (known after apply)
01:35:07.363 STDOUT [app] terraform:       + private_ip                           = (known after apply)
01:35:07.370 STDOUT [app] terraform:       + public_dns                           = (known after apply)
01:35:07.375 STDOUT [app] terraform:       + public_ip                            = (known after apply)
01:35:07.376 STDOUT [app] terraform:       + region                               = "eu-central-1"
01:35:07.378 STDOUT [app] terraform:       + secondary_private_ips                = (known after apply)
01:35:07.379 STDOUT [app] terraform:       + security_groups                      = (known after apply)
01:35:07.380 STDOUT [app] terraform:       + source_dest_check                    = true
01:35:07.381 STDOUT [app] terraform:       + spot_instance_request_id             = (known after apply)
01:35:07.382 STDOUT [app] terraform:       + subnet_id                            = "subnet-123456"
01:35:07.383 STDOUT [app] terraform:       + tags                                 = {
01:35:07.385 STDOUT [app] terraform:           + "Name" = "dev-app"
01:35:07.386 STDOUT [app] terraform:         }
01:35:07.388 STDOUT [app] terraform:       + tags_all                             = {
01:35:07.389 STDOUT [app] terraform:           + "Name" = "dev-app"
01:35:07.391 STDOUT [app] terraform:         }
01:35:07.392 STDOUT [app] terraform:       + tenancy                              = (known after apply)
01:35:07.394 STDOUT [app] terraform:       + user_data_base64                     = (known after apply)
01:35:07.396 STDOUT [app] terraform:       + user_data_replace_on_change          = false
01:35:07.397 STDOUT [app] terraform:       + vpc_security_group_ids               = (known after apply)
01:35:07.403 STDOUT [app] terraform:     }
01:35:07.404 STDOUT [app] terraform: Plan: 1 to add, 0 to change, 0 to destroy.
01:35:07.405 STDOUT [app] terraform:
01:35:07.405 STDOUT [app] terraform: ────────────────────────────────────────────────────────────────────────────────
```

8.Terraform wynik4

```
01:35:07.340 STDOUT [app] terraform:       + host_resource_group_arn              = (known after apply)
01:35:07.341 STDOUT [app] terraform:       + iam_instance_profile                 = (known after apply)
01:35:07.343 STDOUT [app] terraform:       + id                                   = (known after apply)
01:35:07.344 STDOUT [app] terraform:       + instance_initiated_shutdown_behavior = (known after apply)
01:35:07.345 STDOUT [app] terraform:       + instance_lifecycle                   = (known after apply)
01:35:07.346 STDOUT [app] terraform:       + instance_state                       = (known after apply)
01:35:07.346 STDOUT [app] terraform:       + instance_type                        = "t3.micro"
01:35:07.347 STDOUT [app] terraform:       + ipv6_address_count                   = (known after apply)
01:35:07.348 STDOUT [app] terraform:       + ipv6_addresses                       = (known after apply)
01:35:07.349 STDOUT [app] terraform:       + key_name                             = (known after apply)
01:35:07.350 STDOUT [app] terraform:       + monitoring                           = (known after apply)
01:35:07.355 STDOUT [app] terraform:       + outpost_arn                          = (known after apply)
01:35:07.357 STDOUT [app] terraform:       + password_data                        = (known after apply)
01:35:07.358 STDOUT [app] terraform:       + placement_group                      = (known after apply)
01:35:07.359 STDOUT [app] terraform:       + placement_group_id                   = (known after apply)
01:35:07.360 STDOUT [app] terraform:       + placement_partition_number           = (known after apply)
01:35:07.361 STDOUT [app] terraform:       + primary_network_interface_id         = (known after apply)
01:35:07.362 STDOUT [app] terraform:       + private_dns                          = (known after apply)
01:35:07.363 STDOUT [app] terraform:       + private_ip                           = (known after apply)
01:35:07.370 STDOUT [app] terraform:       + public_dns                           = (known after apply)
01:35:07.375 STDOUT [app] terraform:       + public_ip                            = (known after apply)
01:35:07.376 STDOUT [app] terraform:       + region                               = "eu-central-1"
01:35:07.378 STDOUT [app] terraform:       + secondary_private_ips                = (known after apply)
01:35:07.379 STDOUT [app] terraform:       + security_groups                      = (known after apply)
01:35:07.380 STDOUT [app] terraform:       + source_dest_check                    = true
01:35:07.381 STDOUT [app] terraform:       + spot_instance_request_id             = (known after apply)
01:35:07.382 STDOUT [app] terraform:       + subnet_id                            = "subnet-123456"
01:35:07.383 STDOUT [app] terraform:       + tags                                 = {
01:35:07.385 STDOUT [app] terraform:           + "Name" = "dev-app"
01:35:07.386 STDOUT [app] terraform:         }
01:35:07.388 STDOUT [app] terraform:       + tags_all                             = {
01:35:07.389 STDOUT [app] terraform:           + "Name" = "dev-app"
01:35:07.391 STDOUT [app] terraform:         }
01:35:07.392 STDOUT [app] terraform:       + tenancy                              = (known after apply)
01:35:07.394 STDOUT [app] terraform:       + user_data_base64                     = (known after apply)
01:35:07.396 STDOUT [app] terraform:       + user_data_replace_on_change          = false
01:35:07.397 STDOUT [app] terraform:       + vpc_security_group_ids               = (known after apply)
01:35:07.403 STDOUT [app] terraform:     }
01:35:07.404 STDOUT [app] terraform: Plan: 1 to add, 0 to change, 0 to destroy.
01:35:07.405 STDOUT [app] terraform:
01:35:07.405 STDOUT [app] terraform: ────────────────────────────────────────────────────────────────────────────────
01:35:07.408 STDOUT [app] terraform: Note: You didn't use the -out option to save this plan, so Terraform can't
01:35:07.409 STDOUT [app] terraform: guarantee to take exactly these actions if you run "terraform apply" now.

 Run Summary  2 units  22s
 ──────────────────────────
   Succeeded    2
```

9.Terraform-Destroy wynik

```
PS C:\Users\user\Desktop\zad1\lesson34\terragrunt-aws-lab\infrastructure-live\dev> terragrunt run --all destroy
01:40:44.975 WARN   Using `terragrunt.hcl` as the root of Terragrunt configurations is an anti-pattern, and no longer recommende
d. In a future version of Terragrunt, this will result in an error. You are advised to use a differently named file like `root.h
cl` instead. For more information, see https://docs.terragrunt.com/migrate/migrating-from-root-terragrunt-hcl
01:40:45.040 INFO   - Unit app
- Unit vpc

WARNING: Are you sure you want to run `terragrunt destroy` in each unit of the run queue displayed above? There is no undo! (y/n
) y
01:40:50.688 WARN   [app] Config C:\Users\user\Desktop\zad1\lesson34\terragrunt-aws-lab\infrastructure-live\dev\vpc\terragrunt.h
cl is a dependency of C:\Users\user\Desktop\zad1\lesson34\terragrunt-aws-lab\infrastructure-live\dev\app\terragrunt.hcl that has
 no outputs, but mock outputs provided and returning those in dependency output.
01:40:58.275 STDOUT [app] terraform: No changes. No objects need to be destroyed.
01:40:58.284 STDOUT [app] terraform:
01:40:58.284 STDOUT [app] terraform: Either you have not created any objects yet or the existing objects were
01:40:58.285 STDOUT [app] terraform: already deleted outside of Terraform.
01:40:58.288 STDOUT [app] terraform:
01:40:58.289 STDOUT [app] terraform: Destroy complete! Resources: 0 destroyed.
01:40:58.289 STDOUT [app] terraform:
01:41:05.863 STDOUT [vpc] terraform: No changes. No objects need to be destroyed.
01:41:05.864 STDOUT [vpc] terraform:
01:41:05.864 STDOUT [vpc] terraform: Either you have not created any objects yet or the existing objects were
01:41:05.864 STDOUT [vpc] terraform: already deleted outside of Terraform.
01:41:05.877 STDOUT [vpc] terraform:
01:41:05.878 STDOUT [vpc] terraform: Destroy complete! Resources: 0 destroyed.
01:41:05.878 STDOUT [vpc] terraform:

Run Summary  2 units  15s
  ────────────────────────
  Succeeded    2

PS C:\Users\user\Desktop\zad1\lesson34\terragrunt-aws-lab\infrastructure-live\dev>
```

Raport końcowy

Zrealizowano zadanie Multi-Region Deployment przy użyciu Terragrunt.
Utworzono strukturę środowisk dev oraz staging, a także dodano pliki region.hcl.
Skonfigurowano główny plik terragrunt.hcl w celu dynamicznego ustawiania regionu AWS.
Zastosowano blok dependency do przekazywania danych między modułami (vpc → app).
Użyto mock_outputs do poprawnego działania komendy plan bez wcześniejszego apply.
Dane AWS (takie jak klucze dostępu, identyfikatory oraz AMI) zostały ustawione w sposób testowy lub zastępczy, tak aby umożliwić uruchomienie plan bez rzeczywistego tworzenia zasobów w chmurze i bez generowania kosztów.
Wykonano komendę terragrunt run --all destroy.
Terragrunt usuwa zasoby w odwrotnej kolejności zależności: najpierw app, potem vpc, aby zachować spójność infrastruktury.